

RECO API - Complete Endpoint List

1. AUTHENTICATION ENDPOINTS (5 endpoints) (JWT-Based)

POST /api/v1/auth/register

- Create account and automatically login with JWT tokens
- Input: username (string, unique), email (string, unique), password (string, min 8 chars), role (string, optional, default: CUSTOMER)
- Action: Hash password, create USER record, generate access_token + refresh_token
- Returns: {access_token (JWT), refresh_token (JWT), token_type: "Bearer", expires_in: 86400, user: {id, username, email, role, created_at}}
- Status: 201 Created
- Error: 400 Bad Request (validation), 409 Conflict (email exists)

POST /api/v1/auth/login

- Authenticate existing user and return JWT tokens
- Input: email (string), password (string)
- Action: Find user by email, verify password, generate access_token + refresh_token
- Returns: {access_token (JWT), refresh_token (JWT), token_type: "Bearer", expires_in: 86400, user: {id, username, email, role}}
- Status: 200 OK
- Error: 401 Unauthorized (invalid credentials), 404 Not Found (user not found)

GET /api/v1/auth/me

- Get current authenticated user details
- Authorization: Required (Bearer token in header)
- Action: Extract user_id from JWT, fetch user record
- Returns: {id, username, email, role, created_at}
- Status: 200 OK
- Error: 401 Unauthorized (missing/invalid token)

POST /api/v1/auth/refresh

- Get new access token using refresh token (extends session)
- Input: refresh_token (JWT string)
- Action: Verify refresh_token, generate new access_token
- Returns: {access_token (JWT), token_type: "Bearer", expires_in: 86400}
- Status: 200 OK
- Error: 401 Unauthorized (invalid/expired token)

POST /api/v1/auth/logout

- Invalidate user session and blacklist token
 - Authorization: Required (Bearer token in header)
 - Action: Add token to blacklist (database or Redis)
 - Returns: {success: true, message: "Logged out successfully"}
 - Status: 200 OK
 - Error: 401 Unauthorized (missing/invalid token)
-

2. USER ENDPOINTS (4 endpoints)

GET /api/v1/users/{id}

- Get user details
- Authorization: Required (Bearer token)
- Returns: {id, username, email, role, created_at}

PUT /api/v1/users/{id}

- Update user (own data only or ADMIN)
- Authorization: Required (Bearer token)
- Input: username, email, role
- Returns: updated user

DELETE /api/v1/users/{id}

- Delete user (cascades to orders, carts, interactions, cache)
- Authorization: Required (Bearer token)
- Returns: {success: true, message: "User deleted"}

GET /api/v1/users (ADMIN ONLY)

- List all users
- Authorization: Required (Bearer token + ADMIN role)
- Query params: page, limit
- Returns: {total, page, limit, users: [{id, username, email, role, created_at}]}

3. CATEGORY ENDPOINTS (5 endpoints)

GET /api/v1/categories

- List all categories
- Returns: {total, categories: [{id, name, description}]}

GET /api/v1/categories/{id}

- Get category details
- Returns: {id, name, description}

POST /api/v1/categories (ADMIN ONLY)

- Create category
- Input: name, description
- Returns: {id, name, description, created_at}

PUT /api/v1/categories/{id} (ADMIN ONLY)

- Update category
- Input: name, description
- Returns: updated category

DELETE /api/v1/categories/{id} (ADMIN ONLY)

- Delete category
 - Returns: {success: true, message: "Category deleted"}
-

4. PRODUCT ENDPOINTS (6 endpoints)

GET /api/v1/products

- List all products with filters & pagination
- Query params: page, limit, category_id, sort (name/price/popularity), order (asc/desc)
- Returns: {total, page, limit, products: [{id, category_id, name, description, price, stock_quantity, tags, avg_rating}]}

GET /api/v1/products/{id}

- Get product details
- Returns: {id, category_id, name, description, price, stock_quantity, tags, avg_rating, created_at}

POST /api/v1/products (ADMIN ONLY)

- Create product
- Input: category_id, name, description, price, stock_quantity, tags (comma-separated)
- Returns: {id, product details, created_at}

PUT /api/v1/products/{id} (ADMIN ONLY)

- Update product
- Input: category_id, name, description, price, stock_quantity, tags
- Returns: updated product

DELETE /api/v1/products/{id} (ADMIN ONLY)

- Delete product (cascades to carts, orders, interactions, similarities, cache)
- Returns: {success: true, message: "Product deleted"}

GET /api/v1/products/search

- Search products by name/description
- Query params: q (search term), limit
- Returns: {total, products: [{id, name, price, avg_rating}]}

5. CART ENDPOINTS (5 endpoints)

GET /api/v1/users/{user_id}/cart

- Get user's cart
- Returns: {id, user_id, total_price, item_count, items: [{product_id, name, price, quantity}], updated_at}

POST /api/v1/users/{user_id}/cart/items

- Add item to cart
- Input: product_id, quantity
- Returns: {cart_item_id, cart_id, product_id, quantity, total_price}

PUT /api/v1/carts/{cart_id}/items/{cart_item_id}

- Update cart item quantity
- Input: quantity
- Returns: updated cart item with new total_price

DELETE /api/v1/carts/{cart_id}/items/{cart_item_id}

- Remove item from cart
- Returns: {success: true, message: "Item removed"}

DELETE /api/v1/users/{user_id}/cart

- Clear entire cart
 - Returns: {success: true, message: "Cart cleared"}
-

6. ORDER ENDPOINTS (5 endpoints)

GET /api/v1/users/{user_id}/orders

- Get user's orders
- Query params: page, limit, status (Pending/Completed/Cancelled)
- Returns: {total, page, limit, orders: [{id, user_id, total_amount, status, created_at}]}

POST /api/v1/users/{user_id}/orders

- Create order from cart
- Action: Create ORDER + ORDER_ITEMS from CART_ITEMS, then clear cart
- Input: {} (uses existing cart items)
- Returns: {id, user_id, total_amount, status: "Pending", items: [{product_id, quantity, price_at_purchase}], created_at}

GET /api/v1/orders/{id}

- Get order details
- Returns: {id, user_id, total_amount, status, created_at, items: [{product_id, name, quantity, price_at_purchase}]}

PUT /api/v1/orders/{id}/status (ADMIN ONLY)

- Update order status
- Input: status (Pending/Completed/Cancelled)
- Returns: updated order with new status

DELETE /api/v1/orders/{id}

- Delete order (cascades to order_items, payments)
 - Returns: {success: true, message: "Order deleted"}
-

7. ORDER_ITEM ENDPOINTS (2 endpoints)

GET /api/v1/orders/{order_id}/items

- Get all items in order
- Returns: {total, items: [{id, order_id, product_id, quantity, price_at_purchase}]}

GET /api/v1/orders/{order_id}/items/{item_id}

- Get specific order item
 - Returns: {id, order_id, product_id, name, quantity, price_at_purchase}
-

8. PAYMENT ENDPOINTS (3 endpoints)

POST /api/v1/orders/{order_id}/payment

- Process payment for order
- Input: amount, payment_method (MOCK_GATEWAY/CREDIT_CARD), payment_method_details (optional)
- Returns: {id, order_id, amount, status: "Success/Failed", transaction_date}

GET /api/v1/orders/{order_id}/payment

- Get payment details for order
- Returns: {id, order_id, amount, status, payment_method, transaction_date}

GET /api/v1/payments/{payment_id}

- Get payment by ID
 - Returns: payment details
-

9. CART_ITEM ENDPOINTS (2 endpoints)

GET /api/v1/carts/{cart_id}/items

- Get all items in cart
- Returns: {total, items: [{id, cart_id, product_id, quantity}]}

GET /api/v1/carts/{cart_id}/items/{id}

- Get single cart item
 - Returns: {id, cart_id, product_id, name, price, quantity, total_price}
-

10. USER_INTERACTIONS ENDPOINTS (3 endpoints)

POST /api/v1/users/{user_id}/interactions

- Track user interaction with product
- Input: product_id, interaction_type (click, cart_add)
- Action: Increment total_clicks or total_cart_adds in PRODUCTS table
- Stores in: USER_INTERACTIONS table
- Returns: {id, user_id, product_id, interaction_type, created_at}

GET /api/v1/users/{user_id}/interactions

- Get user's interaction history
- Query params: page, limit, interaction_type (click/cart_add)
- Returns: {total, page, limit, interactions: [{id, product_id, interaction_type, created_at}]}

GET /api/v1/products/{product_id}/interactions (ADMIN)

- Get all interactions for a product
 - Returns: {product_id, total_clicks, total_cart_adds, interaction_timeline: [{user_id, interaction_type, created_at}]}
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11. PRODUCT_POPULARITY ENDPOINTS (1 endpoint)

GET /api/v1/products/{product_id}/popularity

- Get product popularity metrics
 - Calculation: $\text{popularity_score} = (\text{total_clicks} \times 0.6) + (\text{total_cart_adds} \times 0.4)$
 - Returns: {product_id, total_clicks, total_cart_adds, popularity_score}
-

12. PRODUCT_SIMILARITIES ENDPOINTS (Content-Based Filtering) (3 endpoints)

GET /api/v1/products/{product_id}/similar

- Get products similar to given product
- Uses: PRODUCT_SIMILARITIES table with min/max normalization
- Query params: limit (default 10)
- Returns: list of {product_id, name, price, similarity_score, avg_rating}

GET /api/v1/similarities (ADMIN)

- Get all product similarities
- Query params: page, limit
- Returns: list of {similarity_id, source_id, similar_id, similarity_score}

GET /api/v1/similarities/{similarity_id}

- Get specific similarity
- Returns: similarity details {id, source_id, similar_id, similarity_score, created_at}

13. REVIEW ENDPOINTS (6 endpoints)

POST /api/v1/products/{product_id}/reviews

- Create review for product
- Input: rating (1-5), comment (optional, max 500 chars)
- Action: Create REVIEW record, update PRODUCT.avg_rating
- Returns: {review_id, user_id, username, product_id, rating, comment, created_at}

GET /api/v1/products/{product_id}/reviews

- Get all reviews for product
- Query params: page, limit, sort (rating/recent)
- Returns: {total, page, limit, avg_rating, rating_distribution: {5: count, 4: count, 3: count, 2: count, 1: count}, reviews: [{review_id, username, rating, comment, created_at}]}

GET /api/v1/users/{user_id}/reviews

- Get all reviews by user
- Query params: page, limit
- Returns: {total, reviews: [{review_id, product_id, product_name, rating, comment, created_at}]}

PUT /api/v1/reviews/{review_id}

- Update own review
- Input: rating, comment
- Action: Update REVIEW record, recalculate PRODUCT.avg_rating
- Returns: {review_id, user_id, product_id, rating, comment, updated_at}

DELETE /api/v1/reviews/{review_id}

- Delete own review
- Action: Delete REVIEW record, recalculate PRODUCT.avg_rating
- Returns: {success: true, message: "Review deleted"}

GET /api/v1/products/{product_id}/rating

- Get product rating summary
- Returns: {product_id, avg_rating, total_reviews, rating_distribution: {5_stars: count, 4_stars: count, 3_stars: count, 2_stars: count, 1_star: count}}

14. RECOMMENDATION_CACHE ENDPOINTS (Personalized Recommendations) (5 endpoints)

GET /api/v1/users/{user_id}/recommendations

- Get personalized recommendations
- Algorithm:
 - STEP 1: Get products user rated 4+ stars from REVIEW table
 - STEP 2: Find similar products using PRODUCT_SIMILARITIES
 - STEP 3: Calculate weighted recommendation score:
$$\text{recommendation_score} = (\text{similarity_score} \times 0.5) + (\text{user_interaction_factor} \times 0.3) + (\text{popularity_score_normalized} \times 0.2)$$
- Where:
 - similarity_score = from PRODUCT_SIMILARITIES (0-1 range)
 - user_interaction_factor = $(\text{user_clicks_on_product} + \text{user_cart_adds_on_product}) / \text{max_user_interactions}$
 - popularity_score_normalized = $\text{popularity_score} / \text{max_popularity_in_db}$
- STEP 4: Exclude products user already bought or rated
- STEP 5: Check cache: If valid → return; If expired → recalculate and cache
- Query params: limit (default 10)
- Returns: {user_id, total_count, recommendations: [{product_id, name, price, avg_rating, recommendation_score}], cached_at, expires_at}

GET /api/v1/recommendations/popular

- Get trending/popular products (Cold Start)
- Uses: Most popular products by $(\text{total_clicks} \times 0.6 + \text{total_cart_adds} \times 0.4)$
- Query params: limit (default 10)
- Returns: {recommendations: [{product_id, name, price, avg_rating, popularity_score, rank}]}

GET /api/v1/recommendations/cache (ADMIN)

- View all cached recommendations
- Query params: page, limit, user_id (optional), expires_before (optional)
- Returns: {total, page, limit, cache_entries: [{cache_id, user_id, product_count, cached_at, expires_at}]}

DELETE /api/v1/recommendations/cache/{cache_id} (ADMIN)

- Delete specific cache entry
- Returns: {success: true, message: "Cache deleted"}

POST /api/v1/recommendations/refresh (ADMIN)

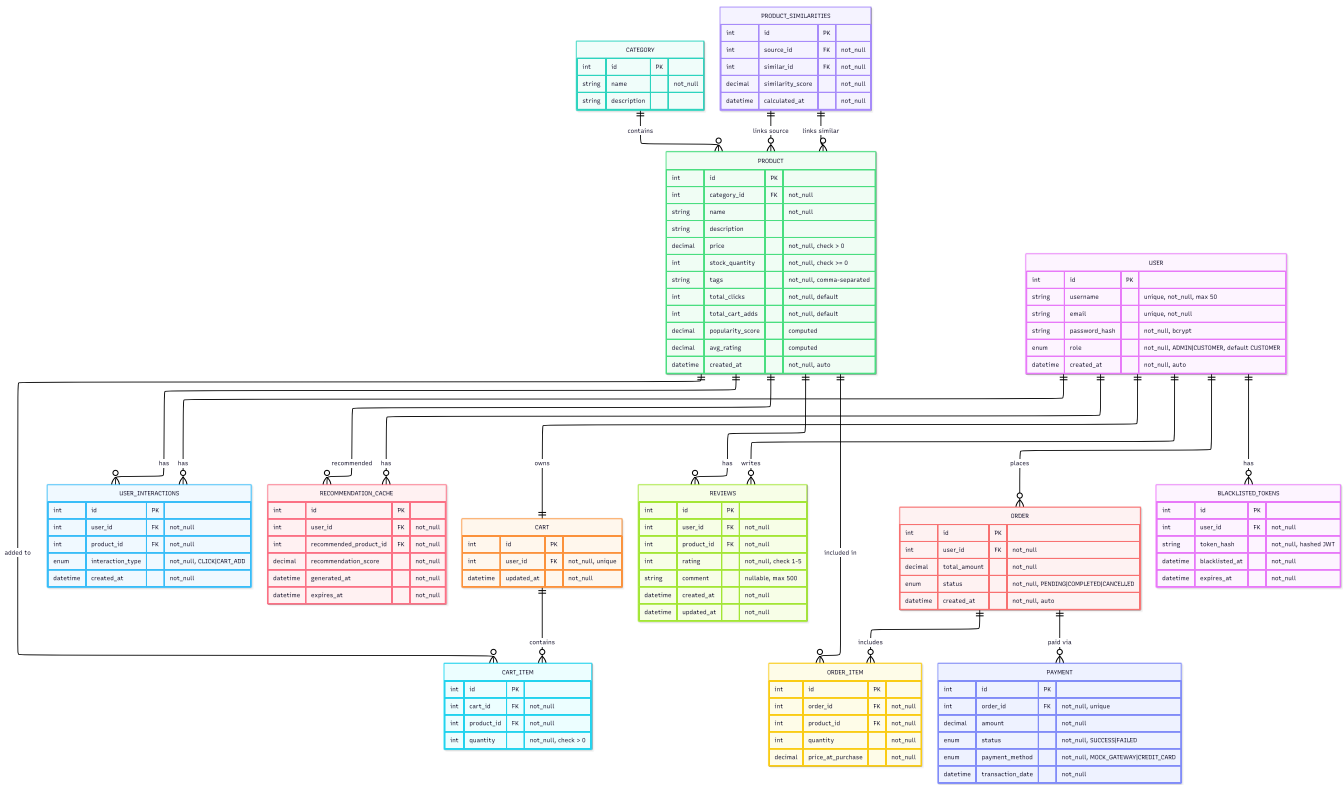
- Manually refresh recommendation cache
 - Input: user_id (optional - if null, refresh all users)
 - Returns: {success: true, users_refreshed: count, cache_entries_created: count}
-

15. FREQUENTLY BOUGHT TOGETHER ENDPOINTS (Upsell) (1 endpoint)

GET /api/v1/products/{product_id}/frequently-bought-together

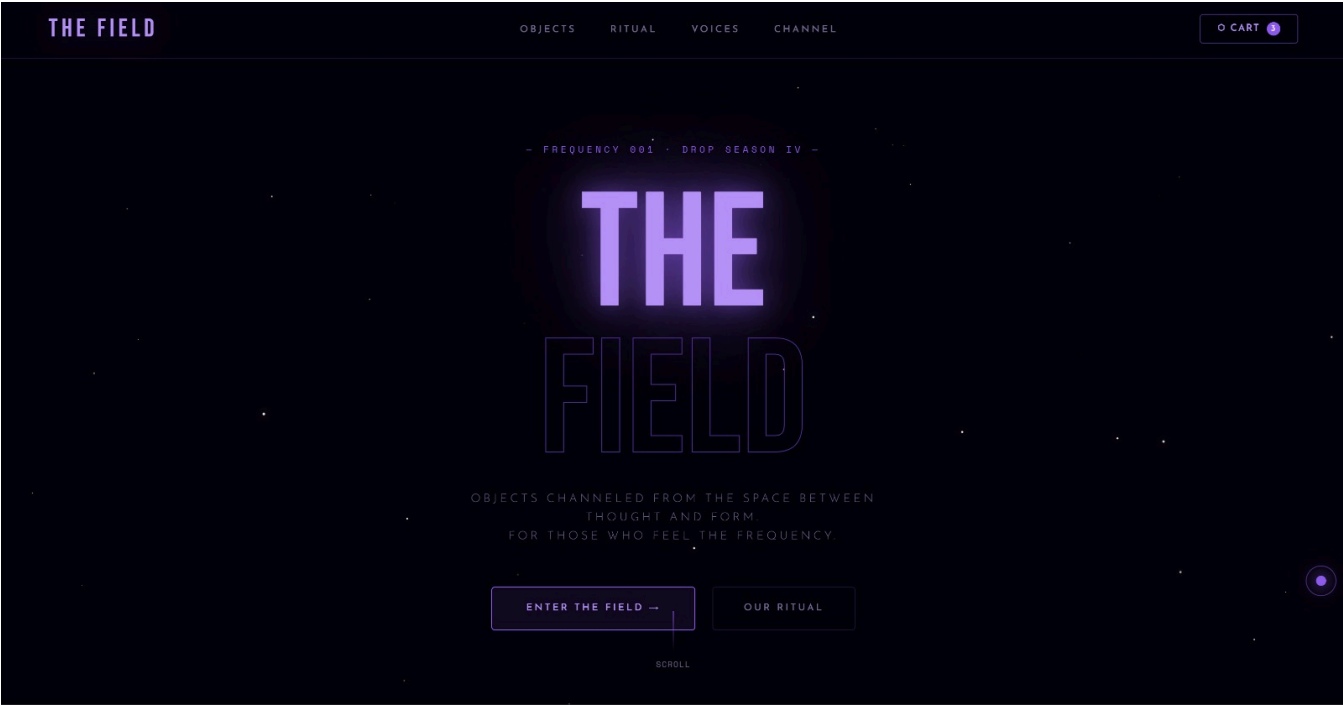
- Get products often bought with this one
- Uses: ORDER_ITEMS (finds co-purchases in same orders)
- Algorithm: Find all orders containing product_id, find other products in those orders, rank by frequency
- Query params: limit (default 5)
- Returns: {product_id, frequently_bought_together: [{id, name, price, frequency, purchase_count, avg_rating}]}

DATABASE SCHEMA

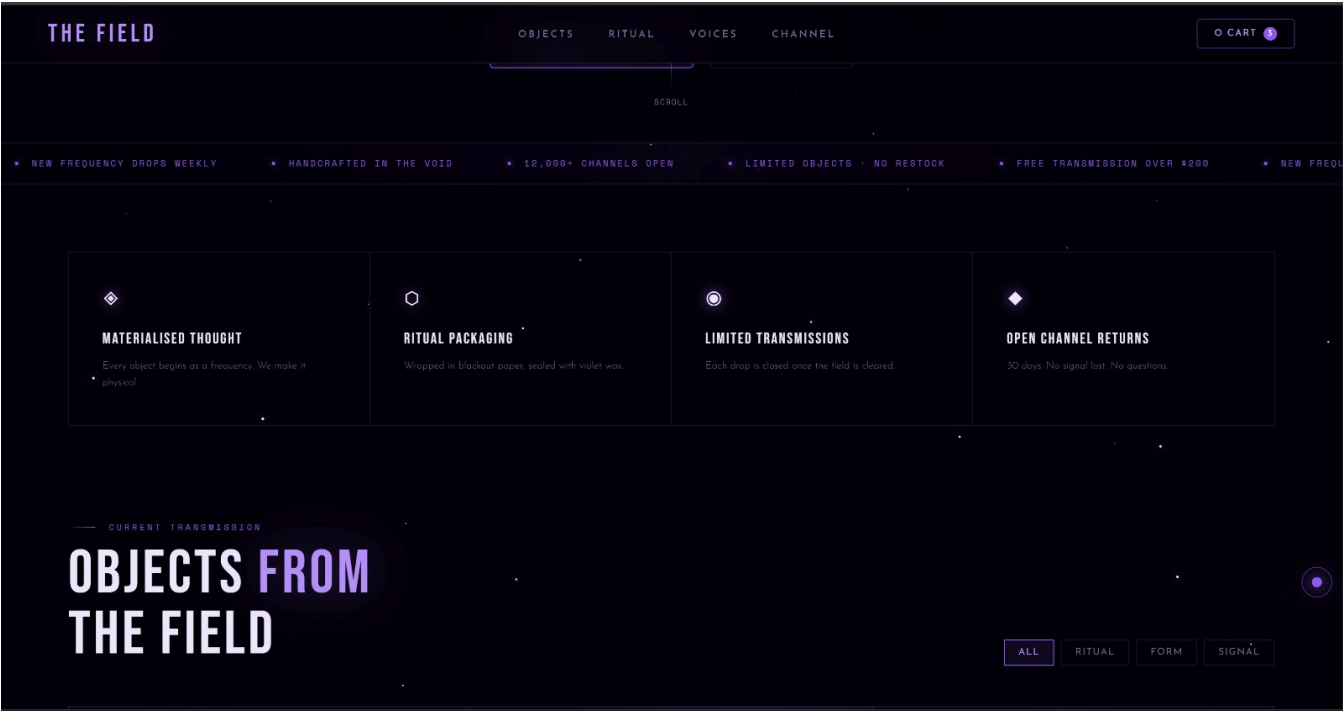


WEBPAGE SCREENSHOTS

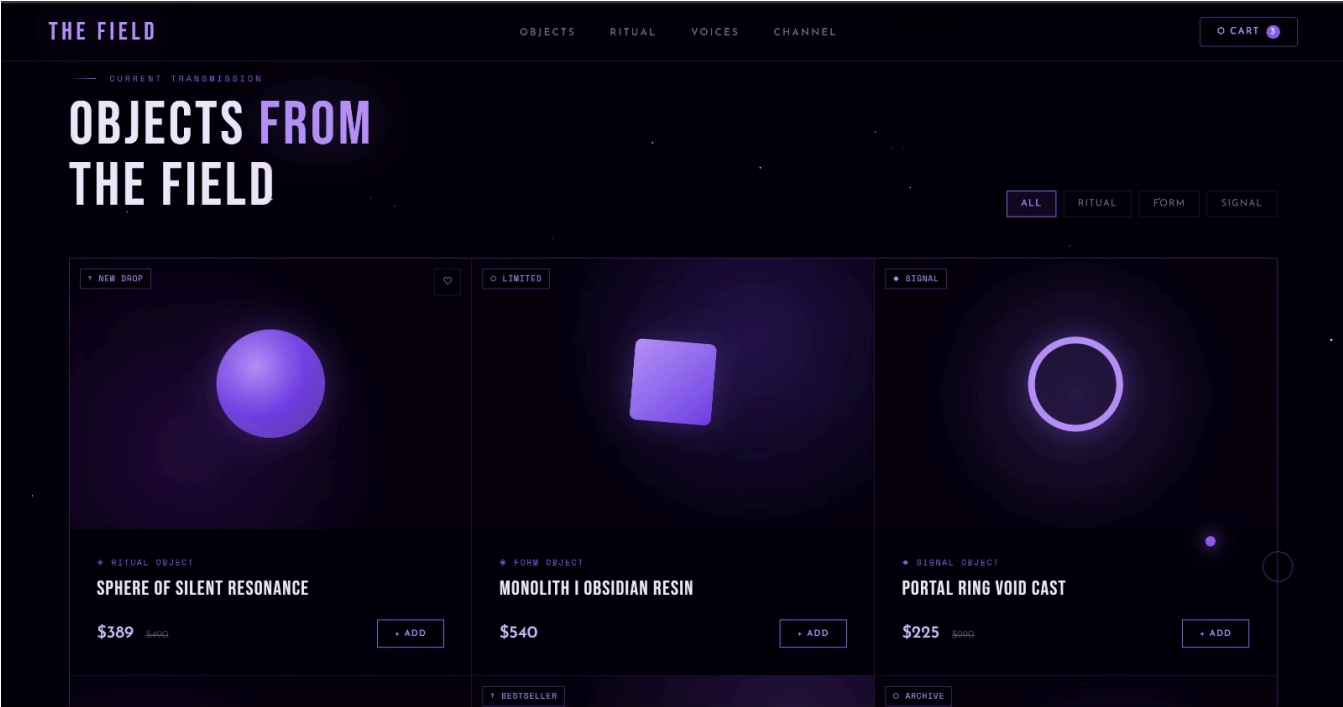
Preview 1



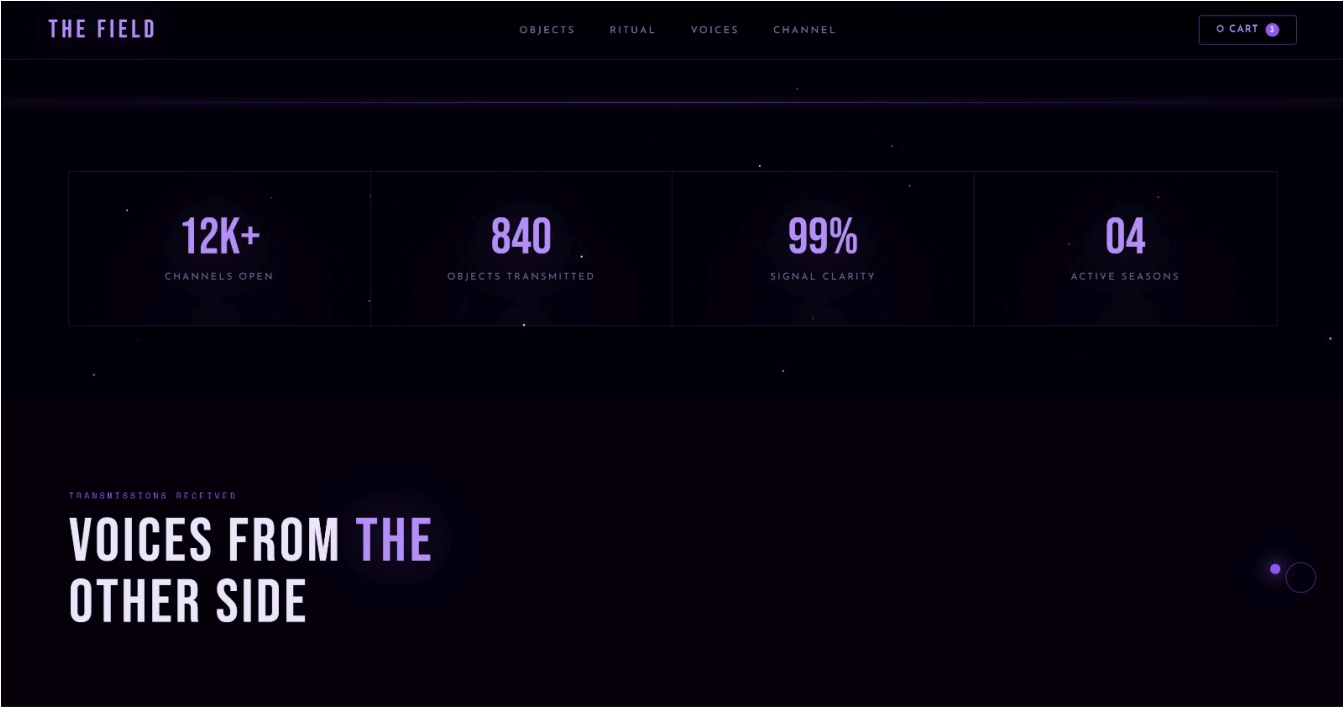
Preview 2



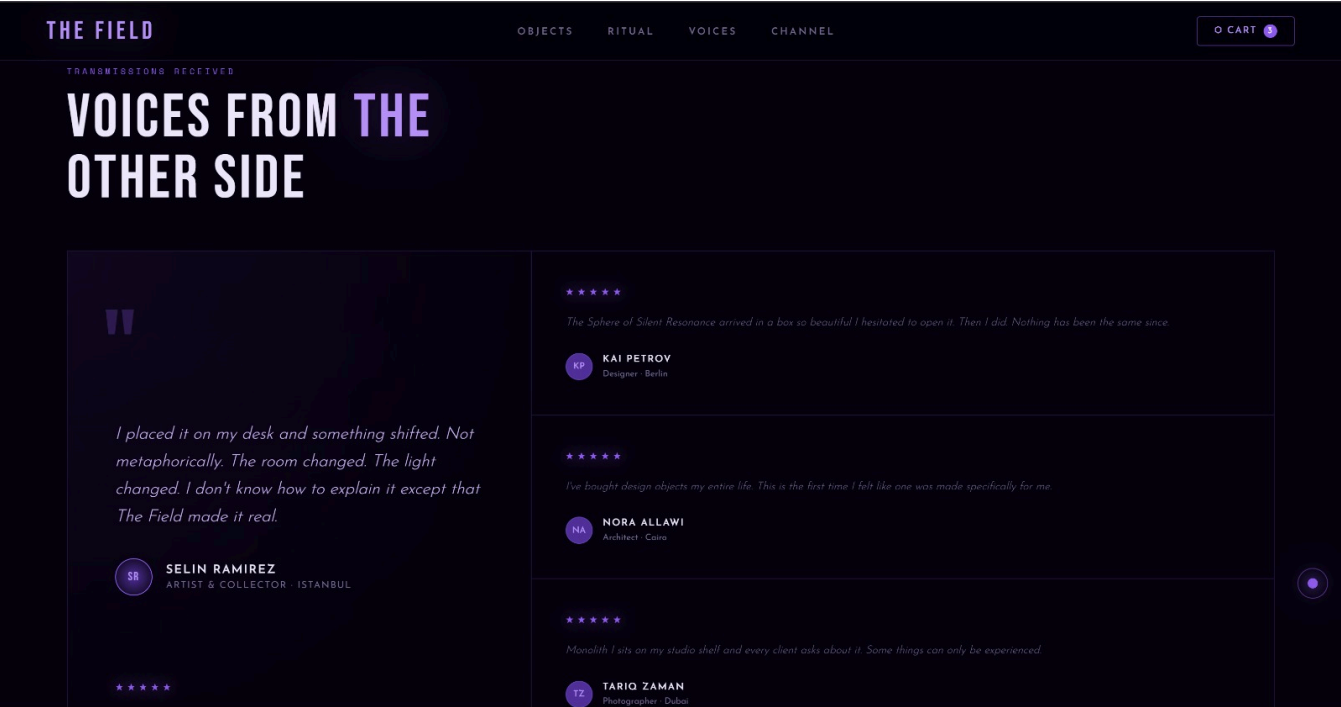
Preview 3



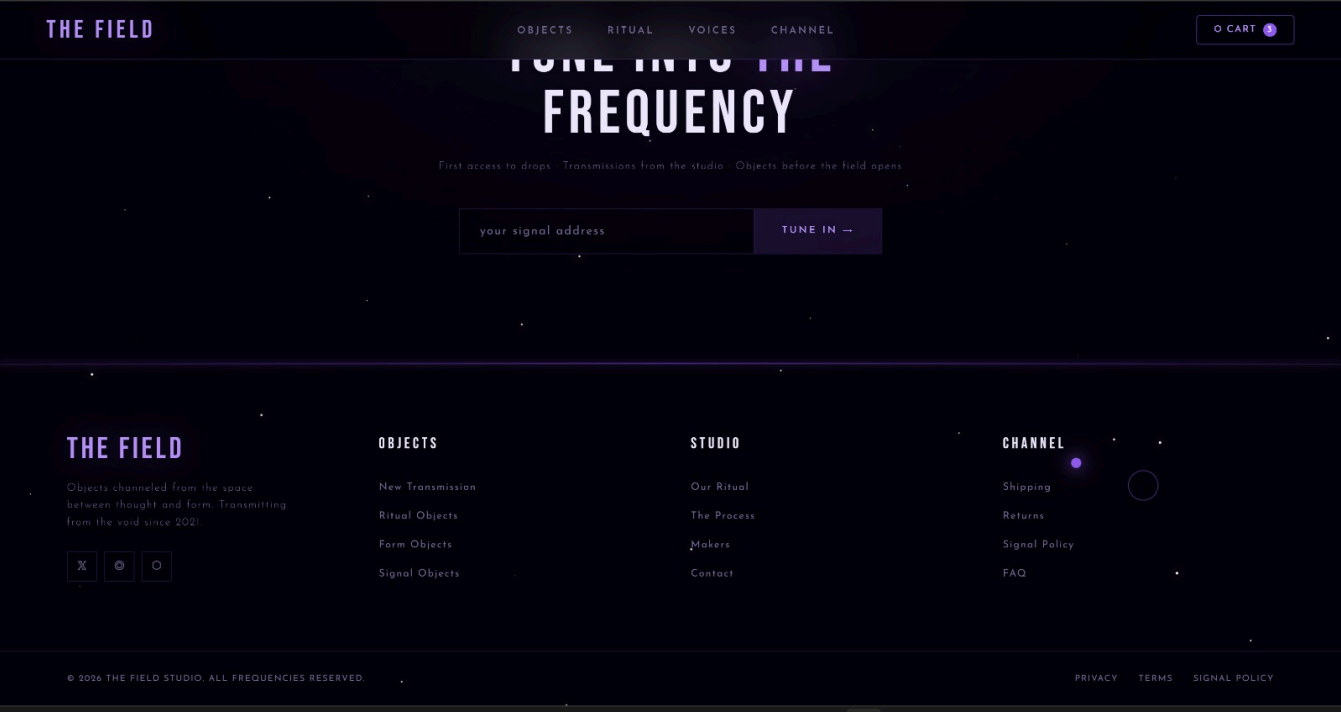
Preview 4



Preview 5



Preview 6



SUMMARY

Total Endpoints: 56

Section	Count
Authentication	5

Section	Count
User	4
Category	5
Product	6
Cart	5
Order	5
Order_Item	2
Payment	3
Cart_Item	2
User_Interactions	3
Product_Popularity	1
Product_Similarities	3
Reviews	6
Recommendations	5
Frequently Bought Together	1